

Descriptive statistics

1. Categorize the following variables as being qualitative or quantitative and specify the level of measurement scale
 - (a) Hair colour: Qualitative; Nominal.
 - (b) Number of television sets in a private home: Quantitative; Ratio.
 - (c) Possible responses to questionnaire items: Qualitative; Ordinal.
 - (d) Intelligence scale of 0 to 100: Quantitative; Interval.
 - (e) The country where you were born in: Qualitative; Nominal.
2. (a) Frequency table using class interval of 1 unit:

Class	[2, 3)	[3, 4)	[4, 5)	[5, 6)	[6, 7)	[7, 8)	[8, 9)
Frequency	13	14	4	4	2	3	1

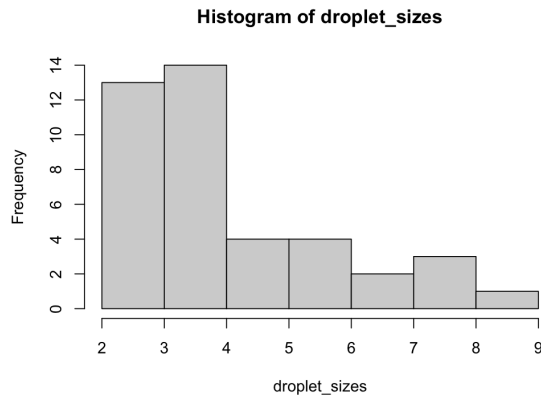


Figure 1: Frequency histogram of droplet sizes.

Shape of distribution: Right-skewed.

(b) Stem-and-leaf diagram

Stem	Leaf
2	1 2 2 3 3 4 5 5 5 8 9 9 9
3	0 1 1 2 3 3 3 4 5 6 6 6 7 7
4	0 2 5 9
5	1 2 3 7
6	0 1
7	1 8 9
8	9

(c)

$$\mu = \frac{\sum X}{N} = \frac{162.6}{41} = 3.97 \text{ (to 2.d.p.)}$$

Procedure for calculating percentiles:

- Compute the rank $R = \frac{P}{100} \times (N + 1)$
- Let I_R = Integer part of R and F_R = Fractional part of R
- P^{th} Percentile =
Data at rank I_R + (Data at rank $(I_R + 1)$ - Data at rank I_R) $\times F_R$

$$25^{\text{th}} \text{ percentile} = 2.8 + (2.9 - 2.8) \times 0.5 = 2.85,$$

since $R = \frac{25}{100} \times (41 + 1) = 10.5$, $I_R = 10$, $F_R = 0.5$ and values are arranged in ascending order.

$$50^{\text{th}} \text{ percentile} = 3.4 + (3.5 - 3.4) \times 0 = 3.4,$$

since $R = \frac{50}{100} \times (41 + 1) = 21$, $I_R = 21$, $F_R = 0$ and values are arranged in ascending order.

$$75^{\text{th}} \text{ percentile} = 4.9 + (5.1 - 4.9) \times 0.5 = 5.0,$$

since $R = \frac{75}{100} \times (41 + 1) = 31.5$, $I_R = 31$, $F_R = 0.5$ and values are arranged in ascending order.

(d) Box plot:

Procedure for calculating box plot:

- Upper Hinge (75^{th} percentile) = 5.0
- Lower Hinge (25^{th} percentile) = 2.85
- $\text{IQR} = 5.0 - 2.85 = 2.15$
- $\text{Step} = 1.5 \times \text{IQR} = 3.225$

- v. Upper Inner Fence = Upper Hinge + 1 Step = 8.225
 $\Rightarrow$ Upper Adjacent (data point just below UIF) = 7.9 and everything beyond that will be denoted as outliers.
- vi. Lower Inner Fence = Upper Hinge - 1 Step = -0.375
 $\Rightarrow$ Lower Adjacent (data point just above LIF) = 2.1 (there are no outliers lower than this)

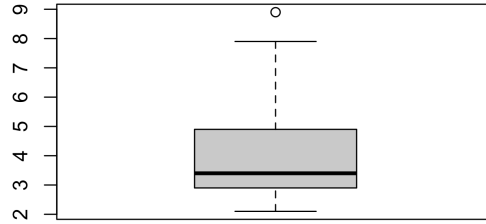


Figure 2: Box plot of droplet sizes.

3. Using the frequency distribution, picture a histogram. Then:

$$\bar{x} = \frac{1 \times 10 + 3 \times 25 + 5 \times 20 + 7 \times 40 + 9 \times 5}{100} = 5.1 \text{ mins}$$

$$\text{mode} = 7,$$

since 7 is the middle of the interval with the highest frequency.

- 4.

$$s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n - 1} = \frac{500 - \frac{50^2}{10}}{10 - 1} = 27.78 \text{ (to 2.d.p.)}$$

$$\Rightarrow s = \sqrt{27.78} = 5.27 \text{ (to 2.d.p.)}$$

If $\sum x^2 = 100$, $s^2 = -16.67$ (to 2.d.p.) but it is not possible to have variance < 0 .

5. Pearson's correlation:

$$\begin{aligned}
\rho &= \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}} \\
&= \frac{437.07 - \frac{26.1 \times 76}{5}}{\sqrt{156.47 - \frac{26.1^2}{5}} \sqrt{1243.42 - \frac{76^2}{5}}} \\
&= \frac{40.35}{\sqrt{20.228} \sqrt{88.22}} \\
&= 0.955 \text{ (to 3.d.p.)}
\end{aligned}$$

X and Y are highly positive correlated.

6. Let A denote the set for the first group of classmates and B denote the set for the second group of classmates. Then:

$$\begin{aligned}
\bar{x}_{A \cup B} &= \frac{\sum_{A \cup B} X}{n_{A \cup B}} \\
&= \frac{\sum_A X + \sum_B X}{n_{A \cup B}} \\
&= \frac{n_A \times \bar{x}_A + n_B \times \bar{x}_B}{(n_A + n_B)} \\
&= \frac{10 \times 2.4 + 5 \times 2.0}{(10 + 5)} \\
&= 2.267 \text{ (to 3.d.p.)}
\end{aligned}$$

From the sample variances, we have:

$$\begin{aligned}
s_A^2 &= \frac{\sum_A X^2 - \frac{(\sum_A X)^2}{n_A}}{n_A - 1} \\
\Rightarrow \sum_A X^2 &= ((n_A - 1) \times s_A^2) + \frac{(\sum_A X)^2}{n_A} \\
&= ((n_A - 1) \times s_A^2) + \frac{(n_A \times \bar{x}_A)^2}{n_A} \\
&= ((10 - 1) \times 0.8^2) + \frac{(10 \times 2.4)^2}{10} \\
&= 63.36
\end{aligned}$$

Similarly:

$$\begin{aligned}\sum_B X^2 &= ((5-1) \times 1.2^2) + \frac{(5 \times 2.0)^2}{5} \\ &= 25.76\end{aligned}$$

Hence:

$$\begin{aligned}s_{A \cup B}^2 &= \frac{\sum_{A \cup B} X^2 - \frac{(\sum_{A \cup B} X)^2}{n_A + n_B}}{n_A + n_B - 1} \\ &= \frac{63.36 + 25.76 - \frac{(10 \times 2.4 + 5 \times 2.0)^2}{10+5}}{10 + 5 - 1} \\ &= \frac{63.36 + 25.76 - 77.06667}{14} \\ &= 0.8609521 \\ \Rightarrow s_{A \cup B} &= \sqrt{0.8609521} = 0.93 \text{ (to 2.d.p.)}\end{aligned}$$